Microsoft Elevate

CAPSTONE PROJECT

Empowering India: A Data-Driven Analysis of Socio-Economic & Health Progress (NFHS-4 vs. NFHS-5)

PRESENTED BY

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OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result (Output Image)
- Conclusion
- Future Scope
- References

PROBLEM STATEMENT:

Despite national growth, there remains a significant disparity in healthcare access, digital literacy, and basic living standards between rural and urban India. Policy makers lack a consolidated, interactive way to track how specific indicators (like the 9.4% Literacy Gap) vary across states over time. This makes it difficult to identify high-risk zones that require immediate infrastructure investment.

PROPOSED SOLUTION:

To bridge the information gap in India’s public health and socio-economic sectors by developing a high-fidelity, interactive "Decision-Support System" using the NFHS-5 dataset.

1. Multidimensional Trend Analysis: A unified interface that transitions seamlessly between three critical development sectors: Digital Empowerment, Maternal Health, and Infrastructure Standards. This allows stakeholders to see how progress in one area (Literacy) affects another (Banking/Mobile access).

2. Granular Comparative Intelligence: Implementation of dynamic slicers and "Rural vs. Urban" toggles. This provides the ability to perform **Comparative Analysis** at a click, revealing hidden disparities that are often lost in national-level averages.

3. Visual Risk Categorization: Instead of raw numbers, the solution uses **Conditional Formatting and Geospatial Logic** to immediately flag "High-Priority" zones. This enables "Management by Exception," where leaders can focus only on the red-flagged regions.

4. Enhanced Navigation UI/UX: An "App-like" navigation framework that ensures non-technical users can explore complex health data without specialized training, making data-driven governance accessible

SYSTEM DEVELOPMENT APPROACH:

1. Data Acquisition (Source):

- Extracted secondary data from the **National Data and Analytics Platform (NDAP)**.
- Dataset includes comprehensive indicators from **NFHS-4 (2015-16)** and **NFHS-5 (2019-21)** for comparative trend analysis

2. Data Preprocessing :

- **Tool** : Power BI Power Query Editor.
- **Cleaning:** Removed null values, handled outliers in mortality data, and standardized state names to ensure geographic accuracy.
- **Transformation**: Performed "Data Pivoting" and changed data types (Text to Decimal) to enable mathematical aggregations.

3. Data Modeling :

- Created a **Star Schema** inspired model by organizing data into logical themes.
- Established relationships between different survey years to allow the "NFHS Version" slicer to function across all visuals.

4. Visualization Design:

- Applied **UI/UX principles** (Consistent color palettes from Coolors, high-contrast text, and intuitive navigation buttons) to ensure the dashboard is accessible by people.

ALGORITHM & DEPLOYMENT:

1. Data Transformation Logic (ETL):

- **Cleaning:** Utilized Power Query to handle missing values and standardize state-level naming conventions across NFHS-4 and NFHS-5.
- **Normalization:** Converted percentage-based strings into decimal/numeric formats to enable mathematical calculations.

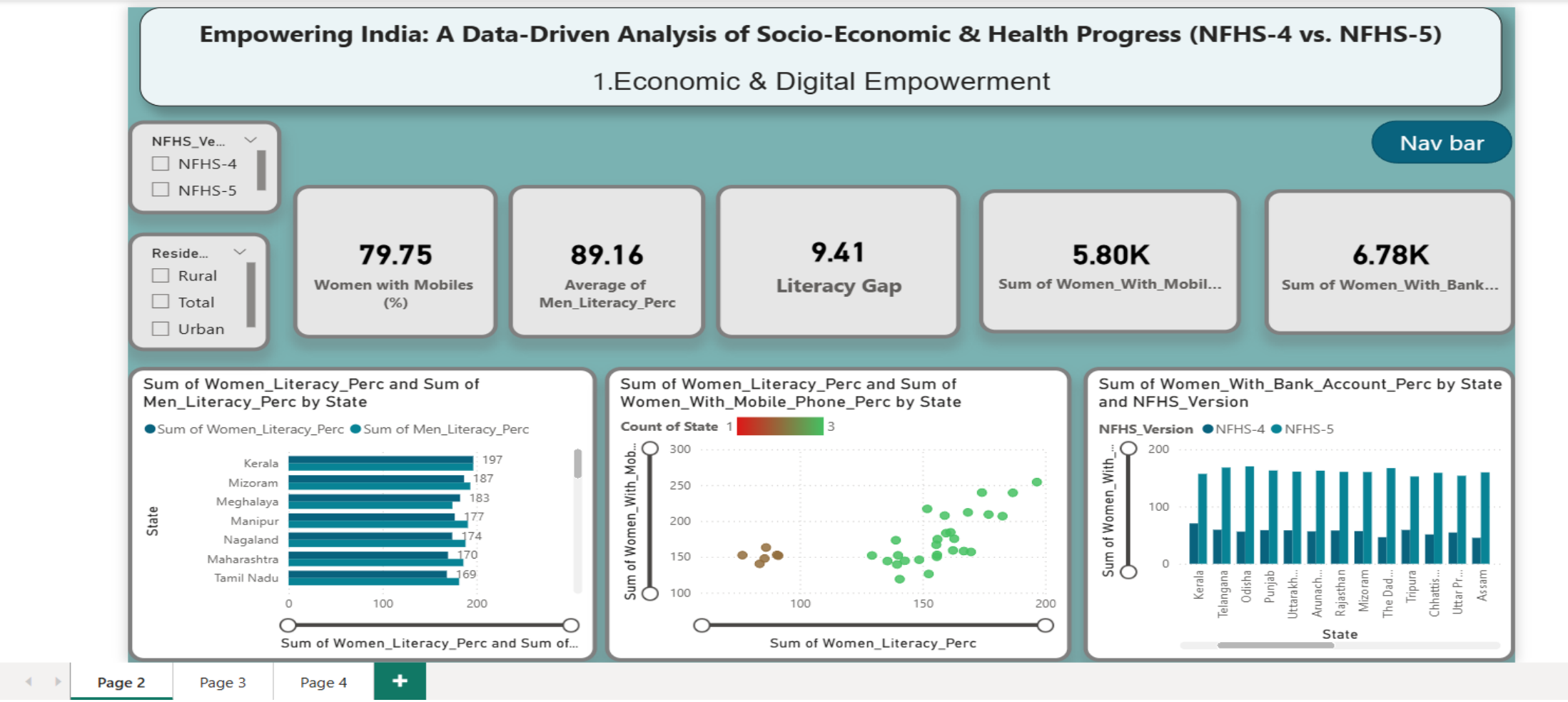
2. Analytical Calculations (DAX Algorithms):

- **Gap Analysis:** Developed a custom measure to calculate the **Gender Literacy Gap** using the formula: Average(Men_Literacy) - Average(Women_Literacy).
- **Risk Categorization:** Implemented a logical "Binning" algorithm. If Infant_Mortality_Rate > National Median, the state is flagged as **"High Risk",** otherwise, it is labeled **"Low/Stable Risk."**

3. Deployment & Interaction:

- **Dynamic Filtering:** Deployed "Slicers" that act as the user interface, allowing real-time filtering by Residence Type (Urban/Rural) and Survey Version.
- **Cross-Highlighting:** Enabled interaction between visuals, where selecting a specific state on one chart automatically filters the health and economic data on the others.

RESULT:



Economic and Digital Empowerment

2.Health & Survival

Maternal & Child Healthcare Outcomes: Identifying High-Risk Zones

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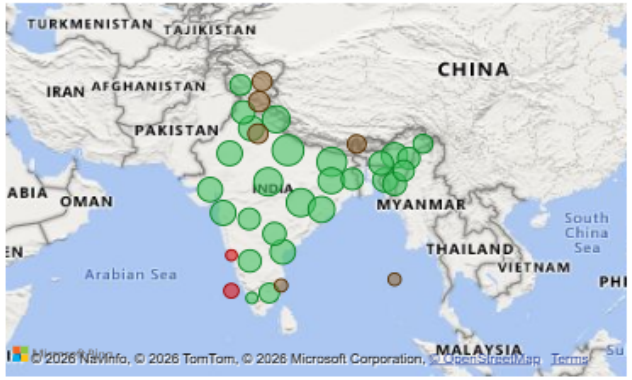
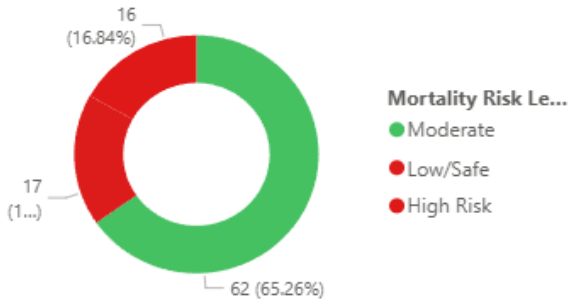
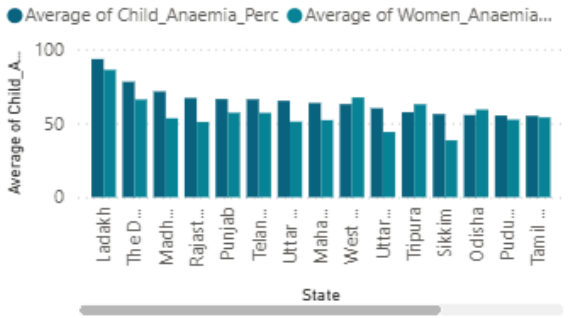
Sum of Infant_Mortality_Rate by State



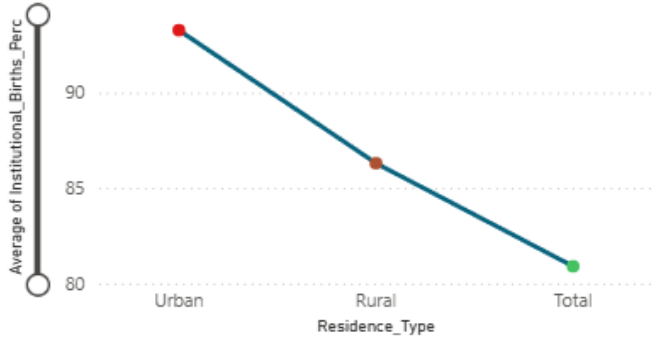
Count of State by Mortality Risk Level



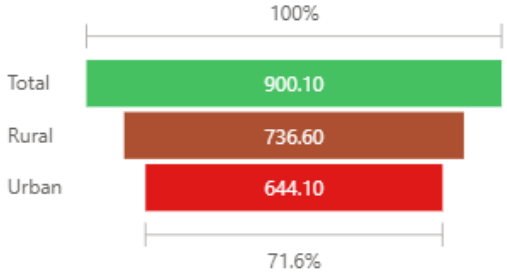
Average of Child_Anaemia_Perc and Average of Women_Anaemia_Perc by State



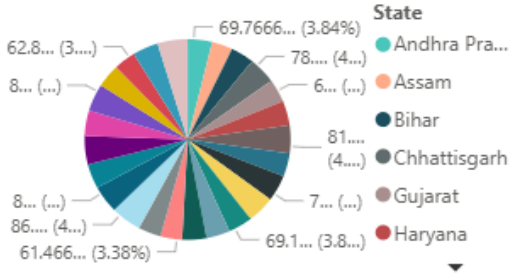
Average of Institutional_Births_Perc by Residence_Type



Sum of Spousal_Violence_Perc by Residence_Type



Average of Child_Full_Vaccination_Perc by State



Health and Survival

3.Quality of Living Standards

Infrastructure & Basic Amenities: Analyzing the Rural-Urban Divide

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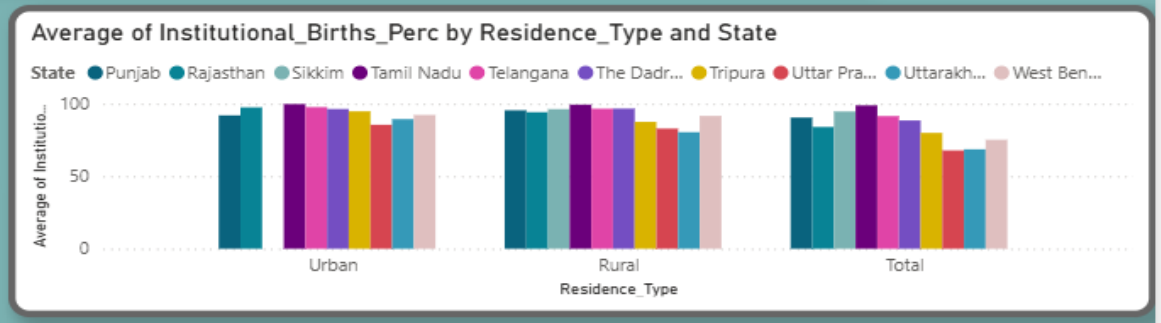
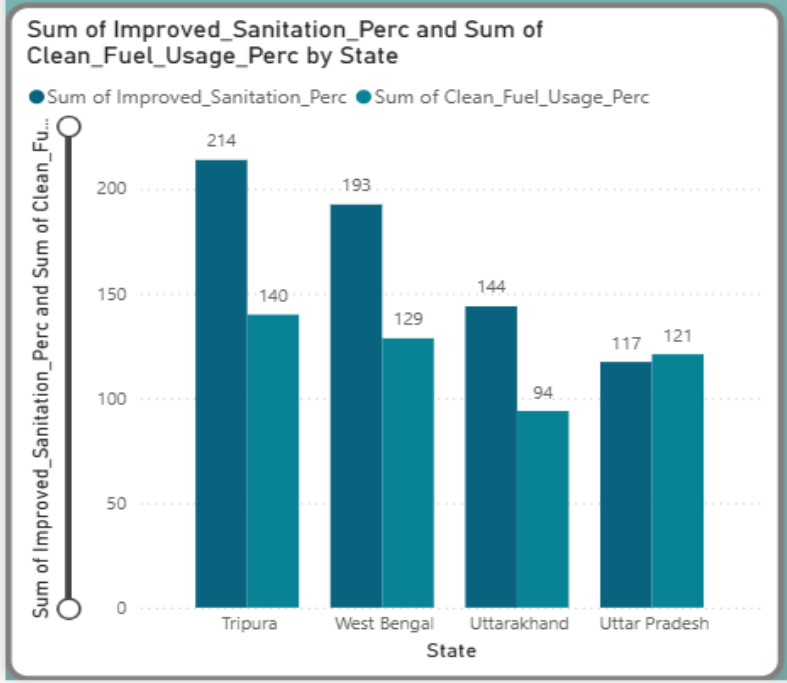
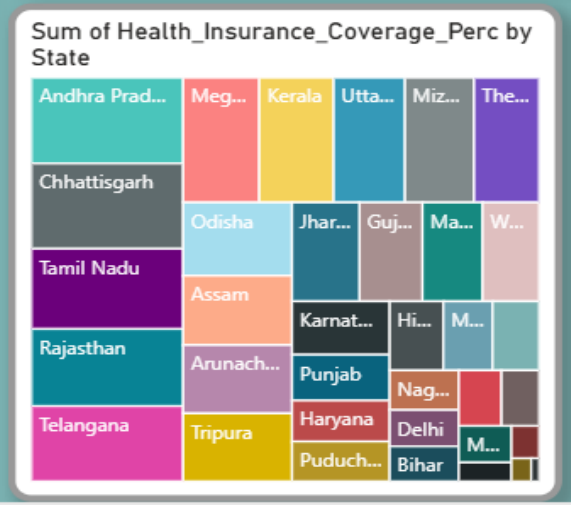
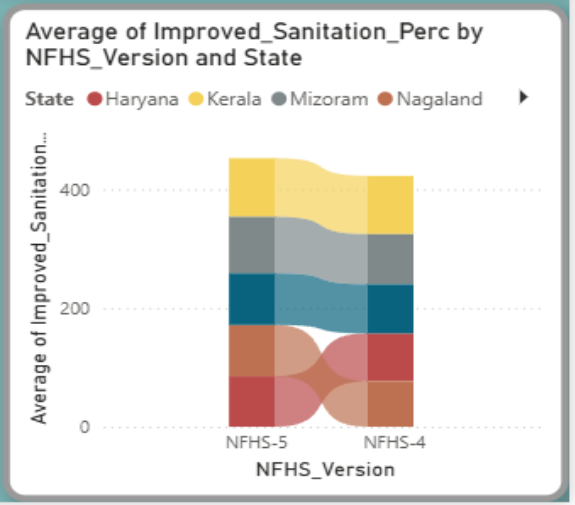
NFHS_Version
- NFHS-4
- NFHS-5









Quality of Living Standards Report

CONCLUSION:

1.The analysis confirms that while India has made strides in financial inclusion, the "Rural-Urban Divide" in infrastructure remains a hurdle.

2.The dashboard successfully provides a data-driven roadmap for targeted development, proving that digital literacy is a major catalyst for overall socio-economic improvement.

FUTURE SCOPE:

1. **Real-time Integration:** Connecting the dashboard to live government APIs for instant data updates.
2. **Predictive Analytics:** Using Machine Learning to forecast future health trends based on current improvement rates.
3. **Granular Mapping:** Expanding the data from State-level to District-level analysis for more local policy impact

REFERENCES:

- Website Link : https://ndap.niti.gov.in/
- Data Source Link: National Data and Analytics Platform (NDAP) - NFHS Dataset
  - **URL:** https://ndap.niti.gov.in/dataset/national-family-health-survey
- GitHub Link: https://github.com/Begginer2/NFHS-Socio-Economic-Dashboard

Thank You